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# Bayesian Networks: Representation

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CS5491: Artificial Intelligence

ZHICHAO LU

Content Credits: [Prof. Wei's CS4486 Course](#)  
and [Prof. Boddeti's AI Course](#)



# TODAY

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Independence

Bayes Net Introduction



# PROBABILISTIC MODELS

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Models describe how (a portion of) the world works



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What do we do with probabilistic models?

- We (or our agents) need to reason about unknown variables, given evidence
- Example: explanation (diagnostic reasoning)
- Example: prediction (causal reasoning)
- Example: value of information



# INDEPENDENCE

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Two variables are *independent* if:

$$\forall x, y : P(x, y) = P(x)P(y)$$



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Independence is a simplifying *modeling assumption*

- *Empirical* joint distributions: at best "close" to independence
- What could we assume for (Weather, Traffic, Cavity, Toothache)?



# EXAMPLE: INDEPENDENCE

---

| T    | W    | P   |
|------|------|-----|
| hot  | sun  | 0.4 |
| hot  | rain | 0.1 |
| cold | sun  | 0.3 |
| cold | rain | 0.2 |



# EXAMPLE: INDEPENDENCE

---

| T    | P   |
|------|-----|
| hot  | 0.5 |
| cold | 0.5 |

| W    | P   |
|------|-----|
| sun  | 0.6 |
| rain | 0.4 |

| T    | W    | P   |
|------|------|-----|
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| T    | W    | P   |
|------|------|-----|
| hot  | sun  | 0.3 |
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| cold | sun  | 0.3 |
| cold | rain | 0.2 |

# EXAMPLE: INDEPENDENCE

---

$N$  fair, independent coin flips:



# EXAMPLE: INDEPENDENCE

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N fair, independent coin flips:

$$P(X_1)$$

|   |     |
|---|-----|
| H | 0.5 |
| T | 0.5 |



# EXAMPLE: INDEPENDENCE

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N fair, independent coin flips:

| $P(X_1)$ |     | $P(X_2)$ |     | $\dots$ |  | $P(X_n)$ |     |
|----------|-----|----------|-----|---------|--|----------|-----|
| H        | 0.5 | H        | 0.5 | $\dots$ |  | H        | 0.5 |
| T        | 0.5 | T        | 0.5 | $\dots$ |  | T        | 0.5 |

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|----------|-----|----------|-----|---------|--|----------|-----|
| H        | 0.5 | H        | 0.5 |         |  | H        | 0.5 |
| T        | 0.5 | T        | 0.5 |         |  | T        | 0.5 |

| $P(X_1, \dots, X_n)$ |          |          |          |  |
|----------------------|----------|----------|----------|--|
| $X_1$                | $X_2$    | $\dots$  | $X_n$    |  |
| H                    | H        | $\dots$  | H        |  |
| H                    | H        | $\dots$  | T        |  |
| $\vdots$             | $\vdots$ | $\vdots$ | $\vdots$ |  |
| T                    | T        | $\dots$  | T        |  |

# CONDITIONAL INDEPENDENCE

---

$P(\text{Toothache}, \text{Cavity}, \text{Catch})$



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If I have a cavity, the probability that the probe catches it does not depend on whether I have a toothache:

- $P(+catch \mid +toothache, +cavity) = P(+catch \mid +cavity)$



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The same independence holds if I do not have a cavity:

- $P(+catch \mid +toothache, -cavity) = P(+catch \mid -cavity)$





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Catch is conditionally independent of Toothache given Cavity:

- $P(catch | toothache, cavity) = P(catch | cavity)$



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Equivalent statements:

- $P(toothache | catch, cavity) = P(toothache | cavity)$
- $P(toothache, catch | cavity) = P(toothache | cavity)P(catch | cavity)$
- One can be derived from the other easily



# CONDITIONAL INDEPENDENCE

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Unconditional (absolute) independence very rare (why?)



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*Conditional independence* is our most basic and robust form of knowledge about uncertain environments.



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X is conditionally independent of Y given Z i.e.,  $X \perp\!\!\!\perp Y | Z$

- iff:

$$\forall x, y, z : P(x, y | z) = P(x | z)P(y | z)$$

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- iff:

$$\forall x, y, z : P(x, y | z) = P(x | z)P(y | z)$$

- or, equivalently, iff

$$\forall x, y, z : P(x | y, z) = P(x | z)$$

# CONDITIONAL INDEPENDENCE

---

Example 1:

- T: Traffic
- U: Umbrella
- R: Raining



# CONDITIONAL INDEPENDENCE

---

Example 1:

- T: Traffic
- U: Umbrella
- R: Raining

Example 2:

- F: Fire
- S: Smoke
- A: Alarm





# CONDITIONAL INDEPENDENCE AND CHAIN RULE

---

Chain rule:  $P(X_1, X_2, \dots, X_n) =$   
 $P(X_1)P(X_2|X_1)P(X_3|X_2, X_1) \dots$



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Trivial decomposition:

$$P(T, R, U) = P(R)P(T|R)P(U|R, T)$$



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With assumption of conditional independence:

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Bayes nets / graphical models help us express conditional independence assumptions



# BAYES NET: BIG PICTURE

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Two problems with using full joint distribution tables as our probabilistic models:

- Unless there are only a few variables, the joint is WAY too big to represent explicitly
- Hard to learn (estimate) anything empirically about more than a few variables at a time



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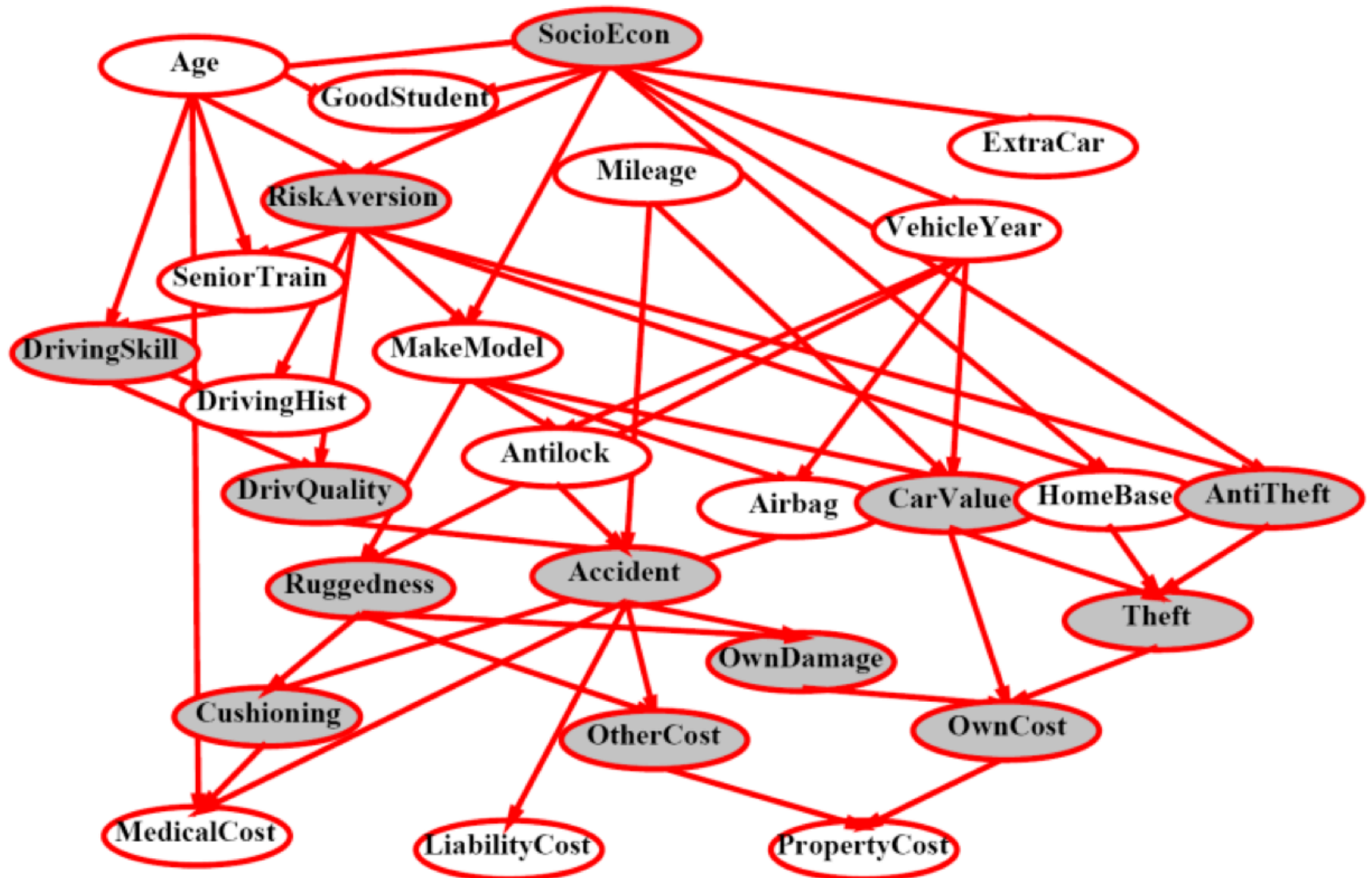
- Unless there are only a few variables, the joint is WAY too big to represent explicitly
- Hard to learn (estimate) anything empirically about more than a few variables at a time

Bayes nets: a technique for describing complex joint distributions (models) using simple, local distributions (conditional probabilities)

- More properly called **probabilistic graphical models**
- We describe how variables locally interact
- Local interactions chain together to give global, indirect interactions

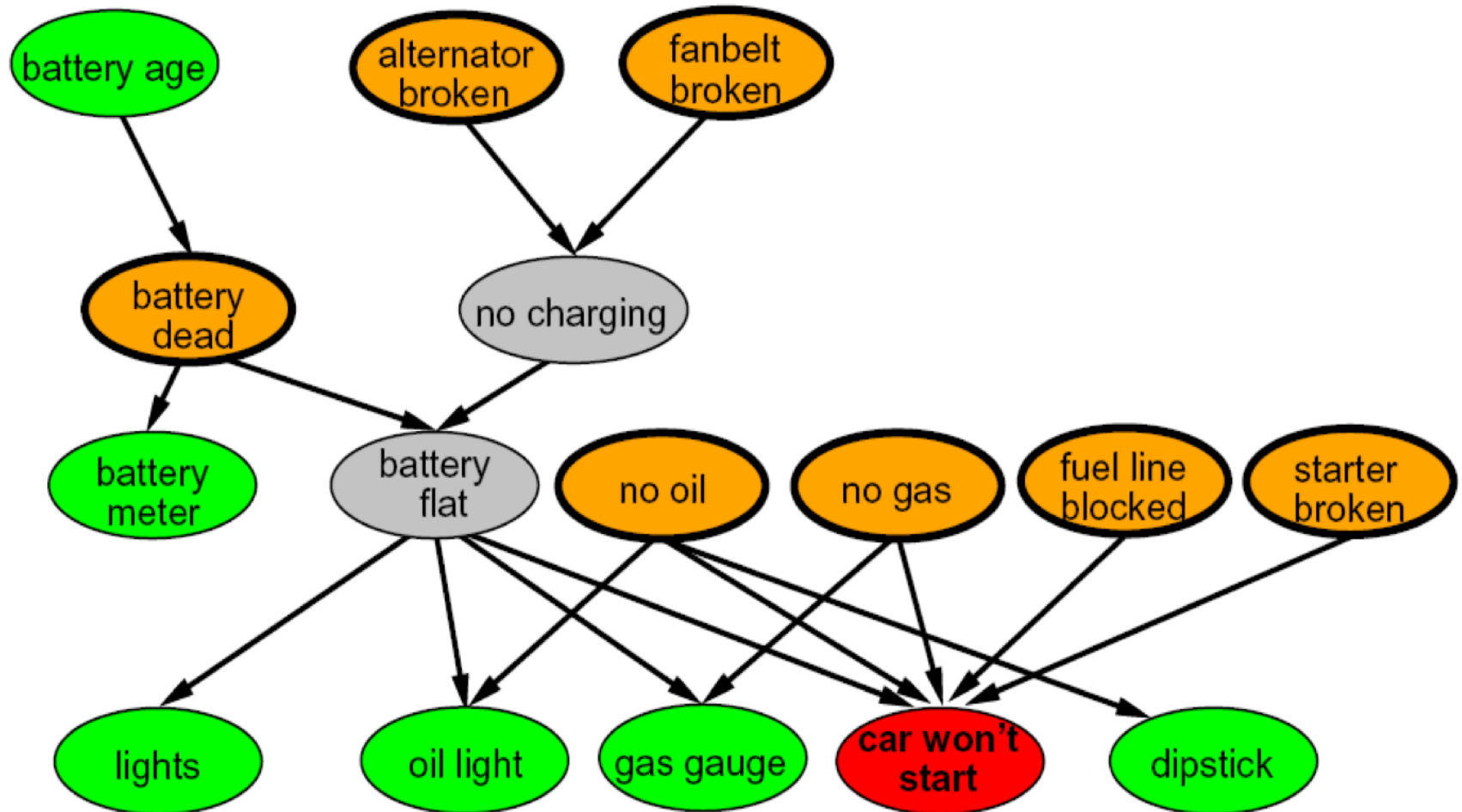


# EXAMPLE BAYES NET: INSURANCE





# EXAMPLE BAYES NET: CAR

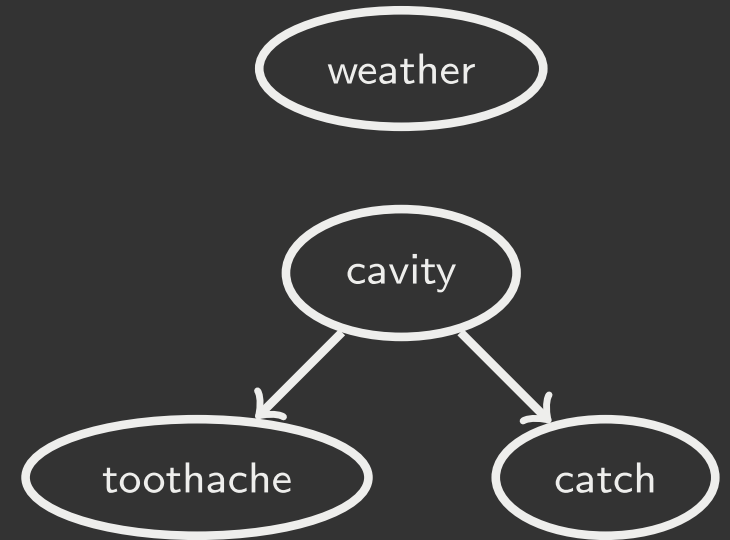


# GRAPHICAL MODEL NOTATION

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Nodes: variables (with domains)

- Can be assigned (observed) or unassigned (unobserved)



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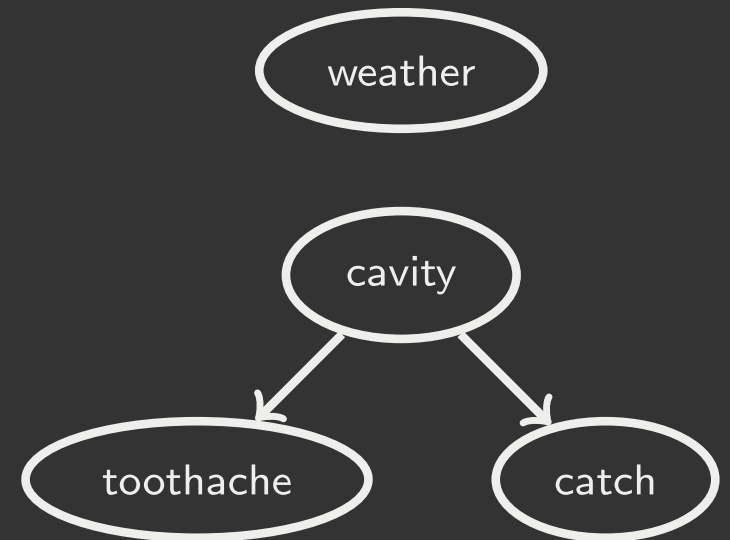
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Arcs: interactions

- Similar to CSP constraints
- Indicate "direct influence" between variables
- Formally: encode conditional independence (more later)



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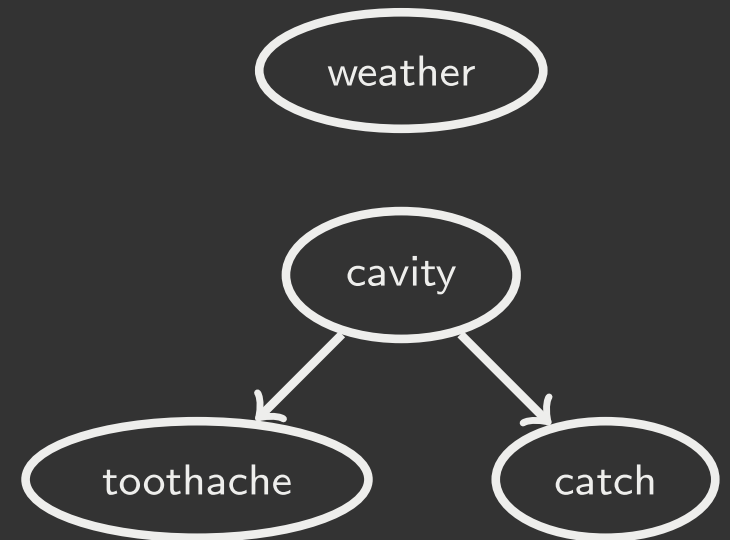
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For now: imagine that arrows mean direct causation (in general, they don't.)



# EXAMPLE: COIN FLIPS

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N independent coin flips



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No interactions between variables: absolute independence



# EXAMPLE: TRAFFIC

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Variables:

- R: it rains
- T: there is traffic



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Model 1: independence





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Model 2: rain causes traffic



# EXAMPLE: TRAFFIC

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- R: it rains
- T: there is traffic

Model 1: independence



Model 2: rain causes traffic



Why is an agent using model 2 better?



# EXAMPLE: TRAFFIC II

---

How do we build a causal graphical model?

Variables:

- T: Traffic
- R: It rains
- L: Low pressure
- D: Roof drips
- B: Ballgame
- C: Cavity



# EXAMPLE: ALARM NETWORK

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How do we build a causal graphical model?

Variables:

- B: Burglary
- A: Alarm goes off
- M: Mary calls
- J: John calls
- E: Earthquake



# BAYES NET SEMANTICS

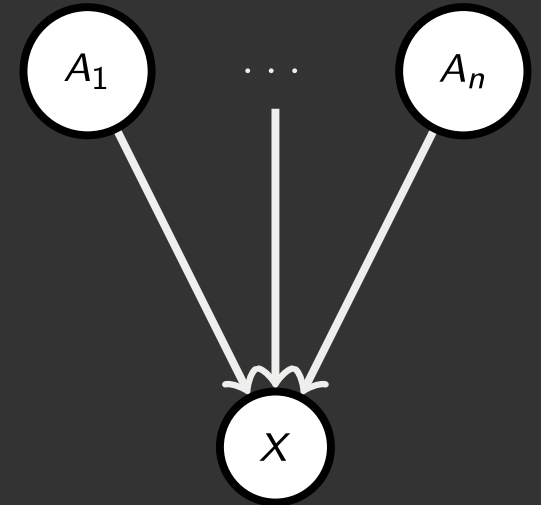
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# BAYES NET SEMANTICS

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A set of nodes, one per variable  $X$



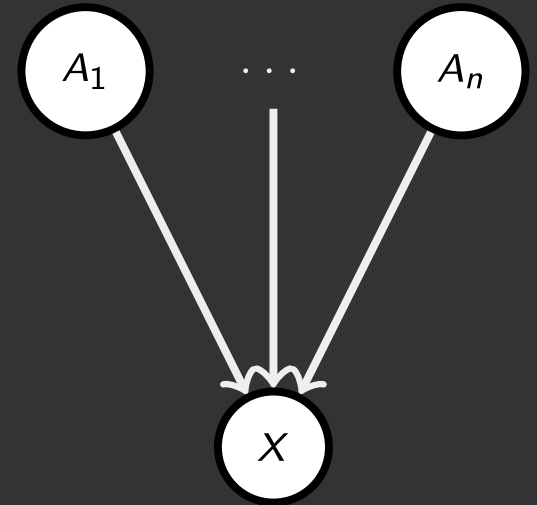
$$P(X|A_1, \dots, A_n)$$

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A directed, acyclic graph



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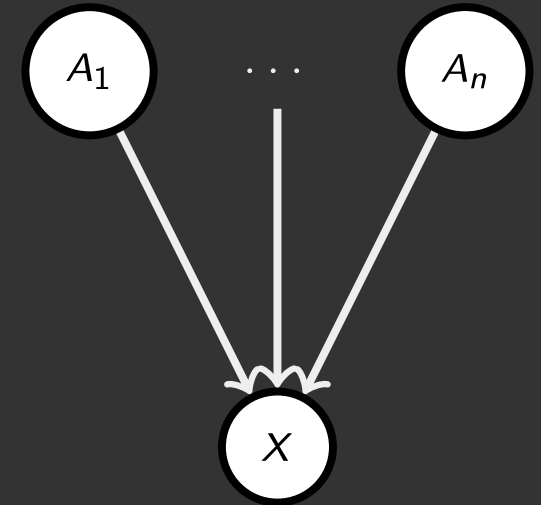
A set of nodes, one per variable  $X$

A directed, acyclic graph

A conditional distribution for each node

- A collection of distributions over  $X$ , one for each combination of parents' values

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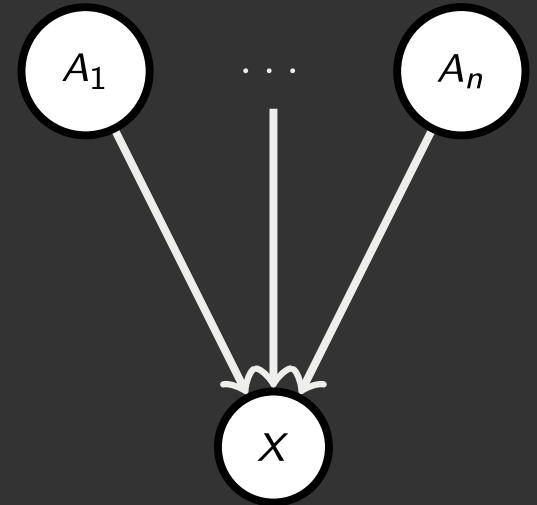
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- CPT: conditional probability table
- Description of a noisy "causal" process



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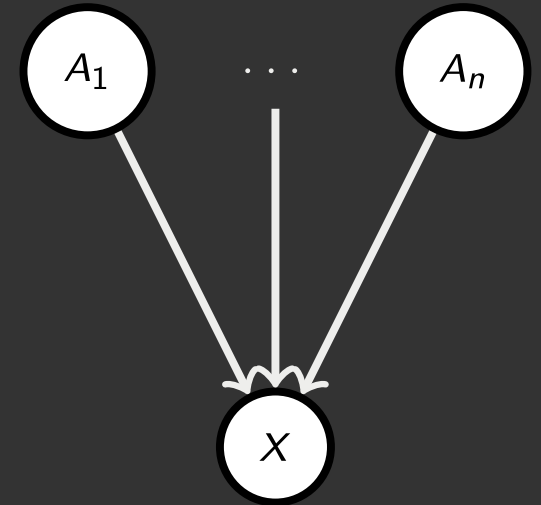
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A Bayes net = Topology (graph) + Local Conditional Probabilities



$$P(X|A_1, \dots, A_n)$$

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Bayes nets **implicitly** encode joint distributions

- As a product of local conditional distributions
- To see what probability a BN gives to a full assignment, multiply all the relevant conditionals together:

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | \textit{parents}(x_i))$$



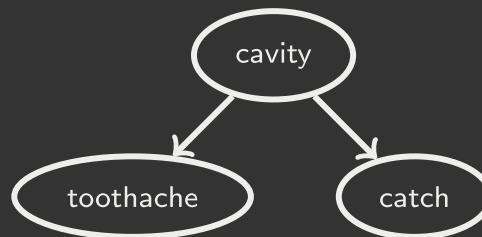
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- Example:  $P(+cavity, +catch, -toothache)$



# PROBABILITIES IN BNS

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Why are we guaranteed that setting the following results in a proper joint distribution?

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Chain rule (valid for all distributions):

$$P(x_1, \dots, x_n) = \prod_{i=1}^n P(x_i | x_1, \dots, x_{i-1})$$





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Not every BN can represent every joint distribution

- The topology enforces certain conditional independencies



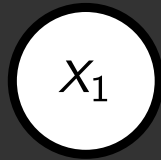
# EXAMPLE: COIN FLIPS

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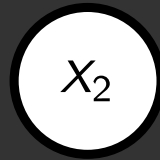
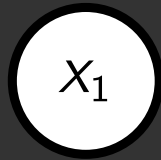


$P(X_1)$

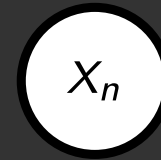
|   |     |
|---|-----|
| h | 0.5 |
| t | 0.5 |

# EXAMPLE: COIN FLIPS

---



...



$P(X_1)$

|   |     |
|---|-----|
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| t | 0.5 |

$P(X_2)$

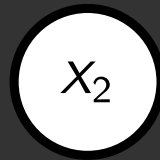
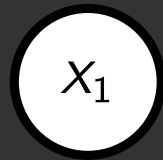
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$P(X_n)$

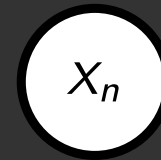
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$P(X_n)$

|   |     |
|---|-----|
| h | 0.5 |
| t | 0.5 |

$$P(h, h, t, h) =$$

Only distributions whose variables are absolutely independent can be represented by a Bayes net with no arcs.

# EXAMPLE: TRAFFIC

---



# EXAMPLE: TRAFFIC

---



$P(R)$

|      |       |
|------|-------|
| $+r$ | $1/4$ |
| $-r$ | $3/4$ |



# EXAMPLE: TRAFFIC

---



$$P(R)$$

|      |       |
|------|-------|
| $+r$ | $1/4$ |
| $-r$ | $3/4$ |

$$P(T|R = +r)$$

|      |       |
|------|-------|
| $+t$ | $3/4$ |
| $-t$ | $1/4$ |

$$P(T|R = -r)$$

|      |       |
|------|-------|
| $+t$ | $1/2$ |
| $-t$ | $1/2$ |

# EXAMPLE: TRAFFIC

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$$P(R)$$

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|      |       |
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| $-t$ | $1/4$ |

$$P(T|R = -r)$$

|      |       |
|------|-------|
| $+t$ | $1/2$ |
| $-t$ | $1/2$ |

$$P(+r, -t) =$$

# EXAMPLE: TRAFFIC

Causal direction



|      | $P(T)$ | $P(R T = +t)$ | $P(R T = -t)$ |
|------|--------|---------------|---------------|
| $+r$ | $1/4$  | $3/4$         | $1/2$         |
| $-r$ | $3/4$  | $1/4$         | $1/2$         |

|      |      |        |
|------|------|--------|
| $+r$ | $+t$ | $3/16$ |
| $+r$ | $-t$ | $1/16$ |
| $-r$ | $+t$ | $6/16$ |
| $-r$ | $-t$ | $6/16$ |

# EXAMPLE: REVERSE TRAFFIC

Reverse Causality?



|      | $P(T)$ |      | $P(R T = +t)$ |      | $P(R T = -t)$ |
|------|--------|------|---------------|------|---------------|
| $+t$ | $9/16$ | $+r$ | $1/3$         | $+r$ | $1/7$         |
| $-t$ | $7/16$ | $-r$ | $2/3$         | $-r$ | $6/7$         |

|      |      |        |
|------|------|--------|
| $+r$ | $+t$ | $3/16$ |
| $+r$ | $-t$ | $1/16$ |
| $-r$ | $+t$ | $6/16$ |
| $-r$ | $-t$ | $6/16$ |

# CAUSALITY

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When Bayes nets reflect the true causal patterns:

- Often simpler (nodes have fewer parents)
- Often easier to think about
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- E.g. consider the variables *Traffic* and *Drips*
- End up with arrows that reflect correlation, not causation



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What do the arrows really mean?

- Topology may happen to encode causal structure
- Topology really encodes conditional independence

$$P(x_i | x_1, \dots, x_{i-1}) = P(x_i | \text{parents}(x_i))$$



# BAYES NET

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So far: how a Bayes net encodes a joint distribution





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Next: how to answer queries about that distribution

- Today:
  - First assembled BNs using an intuitive notion of conditional independence as causality
  - Then saw that key property is conditional independence
- Main goal: answer queries about conditional independence and influence



# BAYES NET

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So far: how a Bayes net encodes a joint distribution

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  - First assembled BNs using an intuitive notion of conditional independence as causality
  - Then saw that key property is conditional independence
- Main goal: answer queries about conditional independence and influence

After that: how to answer numerical queries (inference)



# Q & A



XKCD

